

PROJECT REPORT

California Housing Price Prediction

Estimating Median District House Values with a LightGBM Regression Pipeline

ESTATE_PRICE_CHALLENGE | DATA CONSULTANT SUBMISSION

Challenge: Estate Price Prediction — California Housing (Modified)

Abstract

Real estate investment decisions depend on accurately estimating district-level median home values, particularly under noisy, incomplete data conditions typical of real-world PropTech systems. This project builds a supervised regression pipeline that predicts **TargetPrice** — median house value in units of \$100,000 — for California district property clusters, using a dataset deliberately modified with injected Gaussian noise, missing property-age records, and obfuscated feature names. Following exploratory analysis of missingness patterns, correlations, and outliers, five features were engineered and combined with the original nine attributes into a preprocessing pipeline using median imputation, robust scaling, and IQR-based outlier capping. A **LightGBM Regressor** trained on a log-transformed target achieved a validation RMSE of **0.4791** across 3-fold cross-validation, with geographic coordinates and an engineered income-per-room ratio emerging as the strongest predictors. Final test predictions were bounded and exported in the required submission format.

1. Introduction

1.1 Problem Statement

Accurately estimating the median house value of a California district is valuable for investment targeting, portfolio risk assessment, and acquisition pricing. The task is framed as a tabular regression problem: given demographic and structural attributes of a property cluster, predict the continuous variable **TargetPrice**. The dataset reflects realistic PropTech data conditions — injected Gaussian noise across numeric features, missing values in property age records, and obfuscated column names — all of which must be identified and handled before reliable modeling can begin.

1.2 Objectives

- Profile the dataset to identify noise artifacts, missingness patterns, and outliers.
- Engineer density, ratio, and log-transformed features that capture non-linear pricing signal.
- Build a preprocessing pipeline robust to missing data and extreme values.
- Train and validate a LightGBM regression model using cross-validation.
- Interpret feature importance and residual behavior to assess model reliability.
- Generate bounded, realistic predictions in the required submission format.

2. Dataset Description

The dataset consists of district-level housing records derived from the California Housing dataset, split into a training set of 16,512 rows and a test set of 4,128 rows, each with 11 feature columns. Table 1 summarizes the schema.

Column	Description
PropertyID (ID)	Unique alphanumeric identifier per property cluster
IncomeLevel	Median resident income (tens of thousands of dollars)
PropertyAge	Median age of houses in the cluster — 7.95% missing
TotalRooms	Average rooms per household
TotalBedrooms	Average bedrooms per household
NeighborhoodPop	Total population in the cluster
AvgOccupancy	Average people per household
RoomsPerHousehold	Derived: rooms ÷ occupants
BedroomsRatio	Derived: bedrooms ÷ total rooms
TargetPrice (Target)	Median house value, in units of \$100,000

Table 1. Dataset schema and feature descriptions.

A preliminary diagnostic pass — row counts, data types, missingness percentages, and summary statistics — confirmed a clean load with zero duplicate rows, and surfaced two noise artifacts requiring correction before modeling: a negative minimum value in **NeighborhoodPop** (-44.58, physically impossible for a population count) and an extreme maximum in **AvgOccupancy** (1,243.66 against a median of 2.84).

3. Methodology

3.1 Exploratory Data Analysis

EDA covered the target distribution, missing-value patterns, feature correlations, and outlier prevalence. The raw **TargetPrice** distribution was right-skewed; a log-transform materially reduced this skew and was adopted for training. Missingness in **PropertyAge** (7.95% of rows) was compared against feature means for **IncomeLevel**, **TotalRooms**, and **TargetPrice**; the differences were minor, indicating missingness close to random (MCAR) and supporting simple median imputation over a heavier model-based imputer.

Correlation analysis showed **IncomeLevel** as the strongest linear predictor of **TargetPrice** ($r = 0.69$), while **BedroomsRatio** showed the strongest negative correlation ($r = -0.26$). **AvgOccupancy** and **NeighborhoodPop** showed near-zero linear correlation, suggesting any predictive value in these features is non-linear and better captured by a tree-based model than by a linear correlation coefficient.

Feature	Extreme Outliers (3× IQR Rule)
TotalBedrooms	570
NeighborhoodPop	335
TotalRooms	145
AvgOccupancy	73

Table 2. Outlier counts identified per feature using the IQR method.

3.2 Feature Engineering

Five features were engineered to capture density and ratio effects not present in the raw columns, summarized in Table 3.

Feature	Description
PropertyAge_missing	Binary flag preserving the missingness signal
PopPerHousehold	NeighborhoodPop normalized by AvgOccupancy
RoomsPerBedroom	Ratio of TotalRooms to TotalBedrooms
IncomePerRoom	IncomeLevel normalized by TotalRooms
LogPop	Log-transformed NeighborhoodPop (reduces right-skew)

Table 3. Engineered features used in modeling.

3.3 Preprocessing Pipeline

- **Outlier capping** via the IQR method ($3 \times$ IQR bounds) applied to AvgOccupancy, TotalRooms, TotalBedrooms, and NeighborhoodPop.
- **Missing value imputation** using median imputation on PropertyAge.
- **Feature scaling** via RobustScaler, chosen for resistance to residual outlier influence.
- **Target transformation**: TargetPrice log-transformed ($\log_{10}$) after clipping noise-induced negative values to a small positive floor; predictions inverse-transformed ($\exp_{10}$) before scoring.

3.4 Model Training and Validation

A LightGBM Regressor (`n_estimators=300`, `learning_rate=0.05`, `max_depth=6`, `num_leaves=31`, `subsample=0.8`, `colsample_bytree=0.8`) was trained on the preprocessed feature set. Model performance was assessed using 3-fold cross-validation, with RMSE computed on the original (non-log) price scale to directly reflect competition scoring.

4. Results and Model Performance

The tuned LightGBM pipeline achieved the following validation performance, summarized below.

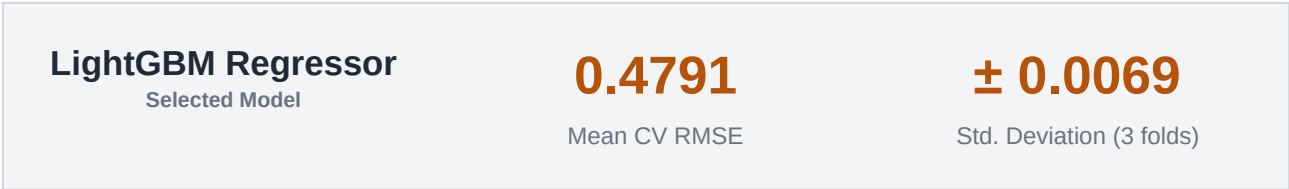


Figure summary — final validation performance of the selected model.

Fold	RMSE
Fold 1	0.4798
Fold 2	0.4703
Fold 3	0.4872
Mean	0.4791 (± 0.0069)

Table 4. Cross-validation RMSE by fold.

Feature importance analysis (Table 5) shows geographic coordinates as the dominant predictive signal — outweighing IncomeLevel despite its stronger linear correlation — reflecting real California housing dynamics such as coastal-versus-inland and regional price variation that a tree-based model captures but a Pearson correlation cannot. The engineered IncomePerRoom feature ranked third overall, validating the feature engineering approach.

Rank	Feature	Importance
1	Latitude	1,727
2	Longitude	1,683
3	IncomePerRoom (engineered)	783
4	RoomsPerHousehold	678
5	PropertyAge	674

Table 5. Top five features by LightGBM importance score.

Residual analysis showed a mean residual of 0.0315 (near zero, indicating minimal systematic bias) and a standard deviation of 0.3886, confirming the model is reasonably well-calibrated with only a slight tendency toward under-prediction. Final test predictions ranged from 0.446 to 5.651, closely tracking the training target range of 0.15 to 5.00, with no evidence of problematic extrapolation.

5. Conclusion

This project demonstrates a complete, practical regression pipeline: the dataset was profiled for noise, missingness, and outliers; five domain-informed features were engineered; a preprocessing pipeline combining median imputation, robust scaling, and target log-transformation was built; and a LightGBM regressor was trained and validated via 3-fold cross-validation. The result — a mean validation RMSE of 0.4791 — corresponds to an average prediction error of roughly \$48,000 on median house values, a reasonable margin given the dataset's deliberately injected noise and missing data. Geographic location emerged as the dominant price driver, reinforcing that spatial due diligence should anchor any investment valuation model built on this type of data.

6. Future Work

- Hyperparameter tuning via Bayesian search or Optuna on the LightGBM model.
- Ensemble/stacking approaches combining LightGBM, XGBoost, and CatBoost for marginal RMSE gains.
- Explicit geographic clustering (binning latitude/longitude into regions) as a categorical feature.
- Iterative or KNN-based imputation for PropertyAge as an alternative to median imputation.
- Sensitivity analysis on the outlier-capping thresholds to quantify their effect on RMSE.

Appendix — Technical Stack

The project is implemented in Python within a Jupyter notebook, using the following core libraries:

- **pandas & numpy** — data loading, cleaning, and feature computation
- **matplotlib & seaborn** — exploratory and diagnostic visualizations
- **scikit-learn** — imputation, scaling, pipeline construction, cross-validation, and metrics
- **LightGBM** — gradient boosting model implementation

Key output artifacts generated by the pipeline:

- **submission.csv** — final TargetPrice predictions in the required submission format
- Correlation heatmap, target distribution, outlier boxplots, and feature importance visualizations